T-T-931C
September 9, 1970
SUPERSEDING
Fed. Spec. T-T-931b
May 24, 1966

FEDERAL SPECIFICATION

TWINE, COTTON: MATTRESS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the requirements for one type of cotton twine (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

FSC 4020

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Military Specification:

MIL-C-3131 - Cordage, Preparation for Delivery Of.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Government or supplier purchases. The requirements specified in 3.6 apply only to twine purchased directly by the Government. All other requirements apply to twine purchased by a supplier as a component for an end item and to twine purchased directly by the Government.

3.2 Materials.

3.2.1 Twine. The twine shall be fabricated from cotton.

3.3 Color. The color of the finished twine shall be tan or natural. The coloration imparted in finishing (see 3.4) will be acceptable.

3.4 Finish. The twine shall have a wax polished finish. The polished twine shall be lustrous and shall have a smooth dressed surface.

3.5 Construction and physical requirements. The finished twine shall conform to the requirements specified in table I and the direction of the final twist shall be "S" when tested as specified in 4.2.5.

TABLE I. Physical requirements

Ply	Length per pound (minimum feet)	Breaking strength (minimum pounds)
6	1575	32

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3.6 Put-up. (see 3.1). The twine shall be put-up in two-pound net weight tubes. A weight tolerance of plus or minus 10 percent shall be allowed on each tube provided that the average net weight does not fall below two pounds. Unless otherwise specified (see 6.2), each tube of twine shall contain not more than a total of two twine knots or splices. The twine shall be continuous throughout and shall be so wound that each turn and layer is free from entanglement.

3.7 Workmanship. The cotton twine shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQLs).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance (DoD). Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection and acceptance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item for visual defects. Defects found during this examination shall be classified in accordance with the defects listed in table II, regardless of their proximity to each other. The sample unit for this examination shall be one tube. Ten percent of the length contained on each sample unit, but not less than 100 feet, shall be subjected to the visual examination. The lot size for this examination

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shall be expressed in units of tubes each. The AQL shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be I.

TABLE II. Visual examination

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Cut, any	X	
	Chafed or damaged	X	
Cleanness	Spot or stain, visible 1/		X
	Lump, picker, or similar dirty waste and linters, visible 1/		X
Color	Not as required	X	
Finish	Not as required	X	

1/ At normal inspection distance (approximately 3 feet).

4.2.3 Examination for weight and winding (see 3.1 and 3.6). The sample unit for this examination shall be one tube. The inspection level shall be S-3 and the AQL shall be 4.0 percent defective. For lots consisting of 500 or fewer units, the sample size (number of sample units) shall be 10 and the acceptance 1. The lot size shall be expressed in units of tubes each. Defects shall be as listed in 4.2.3.1, 4.2.3.2, and 4.2.3.3.

4.2.3.1 Net weight per tube. A defect with regard to weight shall be considered to exist if the net weight of any one tube is less than the minimum or more than the maximum specified in 3.6.

4.2.3.2 Average weight. No AQL is applicable to this examination for average weight. A lot shall be unacceptable if the average weight is less than the two-pound net weight specified in 3.6.

4.2.3.3 Examination for winding. Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

(a) Improperly or not firmly wound resulting in kinks, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of twine.
(b) Put-up not as specified i.e., not tubes.
(c) Any holder containing more than the maximum allowable number of knots or splices (unless otherwise specified see 3.6).

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(d) Not in a continuous length.
(e) Secured other than by knotting or splicing to make a continuous length.

4.2.4 Examination of preparation for delivery requirements. An examination shall be made to determine whether the packaging, packing, and marking complies with the section 5 requirements of this specification. The examination shall be in accordance with the provisions of MIL-C-3131, except that the inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

4.2.5 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable, and as listed in table IV, shall be followed. The physical values specified in section 3 except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test method. The sample size shall be in accordance with table III. The sample unit for testing shall be one tube of twine. The lot size shall be expressed in units of tubes each. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more units fail to meet any requirement. Tests to determine compliance with specification requirements may be made under prevailing atmospheric conditions, except in settlement of dispute, in which case all tests shall be made upon material which has reached equilibrium under standard conditions as defined in Section 4 of Fed. Std. No. 191.

TABLE III. Sampling for tests

	Lot size (tubes)	Sample size
	800 or less	2
	801 up to and including 22,000	3
	22,001 and over	5

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TABLE IV. Test methods

Characteristics	Requirement paragraph	Test method	Number of determinations per individual sample unit	Results reported as:
Material identification	3.2	1200 1/	1	Pass or fail.
Ply	3.5	Visual	1	Pass or fail.
Breaking strength	3.5	4102	5	Average of 5 determinations to nearest 0.5 pound.
Length per pound	3.5	6004	2	Average of 2 determinations to nearest 0.1 foot per pound.
Direction of twist	3.5	4050	3	Pass or fail.

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A or C. Twine, put up as specified, shall be packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B or C. The twine shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking.

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5.3.1 Civil agencies. In addition to any special markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, the interior packages and shipping container shall be marked in accordance with MIL-C-3131.

6. NOTES

6.1 Intended use. The twine covered by this specification is intended for use in tufting or stitching mattresses by hand or machine.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Knots and splices, if other than specified (see 3.6).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

MILITARY CUSTODIANS:

Army - GL
Air Force - 82

Review activity:

Army - MD

User activities:

Navy - SH, SA

Preparing activity:

Army - GL

CIVIL AGENCY INTEREST:

GSA
INT
VA
JUS-FPI

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

T-T-931C: TWINE, COTTON: MATTRESS

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

'A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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